

The Gate Still Does Not Choose: An Expert-Initialised Mixture-of-Experts Beats Its Matched Control and Is the Only All-Rounder Arm under Full-Resolution Cityscapes Metrics

Cityscapes val · ConvNeXt-V2-Base + UPerNet · four experts initialised from the loss specialists B, D, Dp, G · top-2 patch-wise gate over 3×3 patches · 3 seeds × 80 epochs at 1024×2048, against a recipe-matched no-MoE control

Abstract

We report a pre-registered, controlled evaluation of **MoE-V3-CS**, a patch-wise **mixture-of-experts** inserted into a full-resolution Cityscapes segmenter (ConvNeXt-V2-Base + UPerNet, 1024×2048, 19 classes, BF16). Its four experts are **initialised bit-for-bit from the head.fpn_convs.0 block of four independently trained loss specialists of the same seed** — B (CE+Dice), D (CE+Kervadec EDT), Dp (CE+SDT distance map), G (CE+Dice+0.5-blob) — and a top-2 gate over 3×3 patches is then trained for 80 epochs from a residual scale γ initialised to zero, so that epoch 0 is bit-exact to the control. The **pre-registered primary endpoint** is the dataset-level official mIoU (cityscapesScripts, 19 classes) on the shared 500-image val holdout, MoE-V3-CS against its **recipe-matched control** (same B initialisation, same recipe, 80 epochs, no MoE layer), paired image-bootstrap B = 10 000, three seeds averaged within each replicate.

The primary endpoint is **positive at the raw threshold and inside the pre-registered single-pair family**: $\Delta = +0.449$ pt (MoE 81.618 vs control 81.168), 95 % CI [+0.109 ; +0.820], two-sided $p = 0.0066$, Holm = 0.0066 over the 1-pair family. It **does not survive the exploratory multiplicity families**: Holm = 0.0726 over 12 pairs and Holm = 0.0924 over 15 pairs — and **no arm of the plateau survives either** (best 0.0750). All three families are reported together, none is chosen because it flatters. The central result is a **routing diagnostic**: at the final epoch the gate distribution is flat (minimal normalised entropy 0.99771 over the three seeds, part_max 0.5030 / 0.5019 / 0.5019 against 0.5000 for exact equidistribution, every expert between 49.79 % and 50.30 % of the patches, **zero dead experts**), yet γ grows from 0.00024 to ≈ 0.020 ($\times 77$ to $\times 84$) — the mixture **helps**, but as an **average of experts**, not as a selection. The gain does not come from routing; it comes from the expert initialisation. This **replicates on a second dataset, a second architecture and a second attach point** the BRATS result *The Gate Does Not Choose* [DOI 10.5281/zenodo.22903668], of which this paper is a replication, not a republication: no BRATS number is reused.

On the program’s pre-registered versatility criterion (36 endpoints $\times$ 13 arms), MoE-V3-CS is the **only all-rounder**: maximal damage **-0.53 pt** (crowd-group instances $\text{det}@0.5$) against -1.90 (consensus C $\oslash$ B) and -22.20 (D, pedestrian pixel precision 70.0 $\rightarrow$ 47.8), a margin of 1.37 pt over the next arm, and the cheapest plateau of the field at **-1.2 pt of damage per mIoU point** against -43.1 for D. It is nonetheless **first on none of the 36 endpoints** (D wins 16) and its mean percentile is 60.4 (5th of 11): *good everywhere is not best everywhere*. Two business metrics survive the 15-pair Holm — **fragments** (-31.2 connected components per image) and **instances_taille_T3_rappel** (+0.26 pt, largest-instance recall) — while no per-class IoU rise survives the 19-class Holm (the only survivor is a fall, bicycle 0.0038).

Contributions. (1) A pre-registered **positive primary** for an expert-initialised mixture against its recipe-matched control, reported with its **three multiplicity families** together and the explicit statement that it does not survive the exploratory one. (2) A **routing diagnostic that separates a tautology from a proof**: the equality $\text{entropy_token} == \text{entropy_token_clean}$ at the final epoch holds *by construction* (the Shazeer noise is annealed to zero) and proves nothing; the probative comparison runs over the 40 active-noise epochs, where the clean distribution is already flat (≥ 0.99821) and part_max stays within [0.50011 ; 0.50217] of the 0.5000 equidistribution. (3) The correct naming of two routinely conflated routing quantities — $\text{sum}(\text{frac}) == \text{top_k}$ ($= 2$), not 1, and top_expert_share (mean of per-batch maxima, 0.667 at epoch 0) $\neq \text{part_max}$ (maximum of per-batch means, 0.501). (4) A **versatility verdict recomputed from raw deltas and ranks**, with every rank carrying the denominator of its own endpoint (the MoE’s worst rank is 11/12 on **instances_foule_rappel**, because arm A is in partial coverage and the denominator is therefore not constant). (5) A measured **architectural difference with BRATS**: the residual guard γ is indispensable here (without it the raw recipe costs -28.1 mIoU points at epoch 0) and absent there. (6) Public release of code, configs, tables, figures and regeneration scripts.

1. Introduction

A semantic segmenter trained with a single loss is a generalist: it is decent everywhere and excellent nowhere. A natural idea is to build several **specialists** — each trained with a different loss that favours a different behaviour (region overlap, boundary precision, distance regression, per-instance balance) — and then to **combine** them. Two combination routes exist in this program. The first is **training-free consensus**: run the specialists and merge their predictions with a connected-component veto (the C $\oslash$ B / D $\oslash$ B arms).

The second, studied here, is an **internal mixture-of-experts**: insert a patch-wise gate over experts *initialised from the specialists themselves*, and train the gate (and a residual scale) for a short budget while the experts start from real, differentiated weights.

The BRATS track of this program already ran the second route and published a counter-intuitive result, *The Gate Does Not Choose* [DOI 10.5281/zenodo.22903668; concept 10.5281/zenodo.22776410]: the mixture beat its baseline, but the gate distribution stayed **flat** — the win came from *averaging* well-initialised experts, not from the gate *selecting* the right one per patch. This paper asks whether that finding is a property of the BRATS architecture (MedNeXt, 3D, GroupNorm, region-based sigmoid) or a property of the **method**:

On a different dataset, a different architecture, a different probability regime and a different attach point, does an expert-initialised patch-wise mixture still beat its matched control — and does its gate still not choose?

The setting is a controlled arm of a four-paper loss program on Cityscapes at 1024×2048 with ConvNeXt-V2-Base + UPerNet: a distance-map auxiliary regression study [Cassez 2026a, DOI 10.5281/zenodo.21006236], a boundary-loss ablation whose arm B (CE+Dice) supplies both the initialisation and the recipe-matched control used here [Cassez 2026b, DOI 10.5281/zenodo.21006393], a blob-loss study whose arm G serves as expert 3 [Cassez 2026c, DOI 10.5281/zenodo.23083560], and this mixture-of-experts report. The answer, measured before this manuscript was written and pre-registered in the program’s artifact chain (protocol P3.10), is **yes on both counts**: the mixture beats its matched control ($\Delta +0.449$ pt, $p = 0.0066$), and its gate does not choose (flat routing, zero dead experts, $\gamma > 0$). The contribution is therefore not a performance claim — after the exploratory Holm correction no arm of the plateau is significant on mIoU — but a **replication of a mechanism** (average, not selection) and a **versatility diagnostic** (the only arm that is nowhere broken).

2. Related work

Mixture-of-experts. The sparse gate over experts of Shazeer *et al.* [2017] and the top-k routing of Lepikhin *et al.* [2020] establish the load-balanced, noise-injected gate that this paper uses verbatim (switch balancing loss, additive Gaussian noise on the logits annealed over training). Those works scale *transformer* experts and rely on the gate to specialise them. The present setting is the opposite: four *convolutional* experts that already exist, differentiated not by routing pressure but by the **loss they were trained with**, and a short fine-tuning budget. The question is then whether the gate re-specialises them or merely averages them.

Mixture-of-experts from initialised specialists. The BRATS MoE-V3 [Cassez & Larnier 2026, DOI 10.5281/zenodo.22903668] trains a top-2 patch-wise gate over four experts initialised from independently trained specialist arms, and reports that the gate stays flat while the mixture wins. This paper transposes that design to Cityscapes — different dataset (2D urban driving vs 3D glioma), different backbone (ConvNeXt-V2-Base + UPerNet vs MedNeXt), different attach point (`head.fpn_bottleneck` vs `dec_block_0`), different metrics (dataset-level mIoU / boundary / instance recall vs lesion-wise Dice / HD95), different probability regime (exclusive softmax vs region-based sigmoid). It cites the BRATS paper as **anteriority** and as a qualitative comparison point; it reuses **none of its numbers**.

Combining segmenters without training. Ensembling and connected-component consensus are the training-free alternative. Within this program the consensus arms (C $\oslash$ B, D $\oslash$ B, Cp $\oslash$ B, Dp $\oslash$ B) merge two specialists’ predictions with a CC veto that prunes spurious fragments at low mIoU cost [Cassez 2026b]; on BRATS the analogous rule is published in [DOI 10.5281/zenodo.22904810]. §6.3 compares the mixture against these consensus arms on the same 13-arm plateau: two of them (D $\oslash$ B, Dp $\oslash$ B) reach a slightly higher mIoU amplitude than the mixture, but each carries a maximal damage of -8.70 and -4.77 pt respectively, against -0.53 pt for the mixture.

Loss specialists as experts. The four experts of this mixture are exactly the arms studied in the companion papers: CE+Dice [Cassez 2026b], CE+Kervadec boundary/EDT [Kervadec 2019; Cassez 2026b], CE+SDT distance-map regression [Cassez 2026a], and CE+Dice+0.5·blob [Kofler 2023; Cassez 2026c]. Each is a *dominated specialist* on the versatility criterion of §6.6 — excellent on the endpoints its loss targets, broken elsewhere — which is precisely what makes their combination informative: if the mixture merely averaged them it would inherit their breaks; the measurement shows it does not.

3. Method — an expert-initialised patch-wise mixture

Base. ConvNeXt-V2-Base [Woo 2023] pretrained ImageNet-22K (FCMAE) then fine-tuned on ImageNet-1K, + UPerNet head [Xiao 2018], full input resolution 1024×2048 without cropping, 19 Cityscapes classes, BF16 autocast, `channels_last`.

Attach point. A PatchMoE2D layer is inserted as the **last element** of the Sequential `head.fpn_bottleneck` (indices preserved: 0 = conv, 1 = BN, 2 = ReLU, 3 = MoE), so the block computes `fused + fused + experts(fused)`; the residual is carried by

the MoE layer itself (`src/moe/moe_model.py:attach_patch_moe`). The control checkpoint B loads with `strict=False` and the *only* missing keys are those of the MoE layer (verified at run time, otherwise the run aborts).

Experts. Four experts, each initialised from the `head.fpn_convs.0` block of one of the four loss specialists **of the same seed**, all trained 160 epochs: expert 0 $\leftarrow$ B (CE+Dice, which also supplies backbone+head), expert 1 $\leftarrow$ D (CE+Kervadec), expert 2 $\leftarrow$ Dp (CE+SDT DistMap), expert 3 $\leftarrow$ G (CE+Dice+0.5·blob). Every copy is verified **bit-for-bit**: the maximal deviation is **0.0** across the 4 experts $\times$ 3 seeds (12 copies), and each expert is matched to the *same seed* as its run — hence a strict per-seed pairing. The block choice is locked by an audit (`src/moe/AUDIT_DIVERSITE_EXPERTS.md`): the relative Frobenius distance between experts is median **0.538** across methods (min–max 0.422–0.568, all 15 pairs $\geq$ 0.42) against **0.122** for a backbone block, and the specialisation drift is **not reproducible across seeds** (cosine $\approx$ +0.001 to +0.008, noise floor $1/\sqrt{N} \approx$ 0.0007), which is why the pairing is per-seed. The gate is **random** at initialisation (no pre-pairing).

Routing. Top-2 of 4 experts, 3×3 patch grid, convolutional gate (kernel 3, Conv→GELU→Conv 1×1 + average-pool over the grid), **switch** balancing loss at weight 0.001 (weighted auxiliary 0.0020), **Shazeer noise** std $1.0 \times \text{std}(\text{logits})$ **annealed over 40 epochs**, residual scale initialised to **0**. Exact equidistribution of `part_max` for top-2 of 4 experts is **0.5000** ($= \text{top}_k / n_{\text{experts}}$, computed, never hard-coded); it is the reference against which §6.2 reads the routing.

Residual guard γ (a measured difference with BRATS). Without a residual scale, the raw BRATS recipe costs **−28.1 mIoU points at epoch 0** on this architecture (pixel agreement 91.5 %, truck −87, rider −78, traffic light −70; measured in P3.08), because the experts initialised from `head.fpn_convs.0` receive at the attach point a distribution they have never seen (the output of `fpn_bottleneck`), with no renormalisation before the classifier — a problem absent on BRATS (identical MedNeXt blocks everywhere + GroupNorm). With $\gamma = 0$ **initial, learnable per channel**, epoch 0 is **bit-exact to the control B**, and $|\gamma|$ becomes a direct diagnostic of the thesis ($|\gamma| \rightarrow 0 =$ the mixture is useless). Measured: $|\gamma| \approx$ 0.020 at the end of training (§6.2).

Matched control. `control_nomoe_Binit`: same B initialisation, same recipe, **80 epochs**, *without* the MoE layer (3 seeds). This is the reference of the primary endpoint — not B (160 epochs), not the unmatched control.

Recipe (mixture and control). 80 epochs, batch $2 \times$ gradient accumulation 4 (effective 8), AdamW ($\text{lr} 6 \times 10^{-5}$, weight decay 0.01, betas (0.9, 0.999)), polynomial decay (power 1.0, 1-epoch warmup), BF16, horizontal flip $p = 0.5$.

Two routing quantities, correctly named. `frac` is the share of *patches* processed by each expert (`onehot.clamp_max(1).mean(0)`, `src/moe/patch_moe2d.py`); because each patch activates `top_k` experts, `sum(frac) == top_k (= 2.0)`, **not 1**. `top_expert_share` is the **mean of the per-batch maxima** (the average, over batches, of the largest expert’s share) whereas `part_max` is the **maximum of the per-batch means** (the largest expert’s share averaged over batches); by Jensen $\text{top_expert_share} \geq \text{part_max}$ (measured 0.667 vs 0.501 at epoch 0). Neither is “the share of patches handled by the first expert”; conflating them would make a flat gate look like a choosing one.

4. Experimental setup

4.1 Data, holdout, metrics

Cityscapes fine annotations [Cordts 2016]: 2 975 train / 500 val images, 19 evaluation classes. All evaluations run on the program’s **pre-specified shared holdout**: the first 500 val images in loader order (`first:500`) — identical for all 13 arms, fixed before any arm was trained; it is *not* a random draw. The official Cityscapes leaderboard was **not** submitted to (declared limitation, §8).

- **mIoU** — dataset-level, from a single aggregated confusion matrix, estimator validated **bit-identical to the official cityscapesScripts** routine (`tests/test_official_miou.py`). Per-class IoU from the same matrices.
- **Boundary F1 (3 px)** — per-class F1 of predicted vs ground-truth contours within a 3-pixel tolerance [protocol of Perazzi 2016]; also restricted to {person, rider}.
- **Pedestrian instance metrics** — from the official `*_gtFine_instanceIds.png` maps, classes {person, rider}: strict recall, `detection@0.5`, pedestrian pixel precision; **strata** = all / individual (`instId` $\geq$ 1000) / crowd-group (`instId` < 1000) instances, and **size terciles** T1 (smallest) / T2 / T3 (largest) of individual instances.
- **Fragments** — 8-connected components per image summed over the 19 class masks (`count_fragments`, `src/postprocessing/conse`).

4.2 Pre-registered primary endpoint and statistics

The single pre-registered primary endpoint is the **dataset-level official mIoU, MoE-V3-CS vs its recipe-matched control** (`moe_v3cs` vs `controle`). Test: **paired image-bootstrap** over the 500 holdout images — positions resampled with replacement, **B = 10 000** replicates, bootstrap seed **20260618**, the three seeds (42, 123, 456) averaged *within each replicate*, 95 % percentile CI, two-sided $p = 2 \cdot \min(\text{frac } \Delta \leq 0, \text{frac } \Delta \geq 0)$. The criterion was declared before the analysis runs (protocol P3.10). The primary family is the single pre-registered pair, so Holm = p there.

Three multiplicity families, all reported. The same contrast belongs to three different Holm families depending on which protocol declares it: the single pre-registered pair (P3.10, size 1), the master table of arms-against-control (P3.14, size 12), and the program’s exploratory family that adds the four fusion-versus-expert comparisons (P3.16, size 15). Family sizes are **computed** from the artifacts (`len(pairwise)`), and every Holm value is **recomputed** from the raw p-values then compared against the stored one (agreement $< 1 \times 10^{-12}$, blocking check). This is the house discipline inherited from the paper4 erratum v1.1.0, where a hand-written family label ended up mislabelling the values it carried.

Two significance criteria, never conflated. *Holm-significant* (corrected $p < 0.05$ within the stated family) and *CI-excluding-zero* (raw 95 % CI) are reported **separately** everywhere; bold in tables marks Holm significance only. This matters here: e.g. truck +4.24 IoU excludes zero but does not survive the 19-class Holm (0.1188).

Non-determinism, declared. cuDNN benchmark is on (`deterministic: false`, BF16). The primary numbers come from the harness forward of the pre-registered artifact (P3.10); the business-metric regeneration (§6.4) comes from a second independent GPU forward of the *same* checkpoints. The two agree within **8.58×10^{-4} pt** on the primary Δ (+0.449360 vs +0.450218), of the same order as the 4.13×10^{-3} pt already documented in the program’s consolidation sanity; the verdict is identical in both.

5. Results

5.1 Primary endpoint: positive raw, not surviving the exploratory family

Pre-registered primary, paired image-bootstrap over the 500-image holdout (artifact `results/moe_v3_cs/harness/table_controle_v3cs`)

Quantity	MoE-V3-CS	matched control
mIoU (point)	81.618	81.168
95 % CI	[80.264 ; 82.709]	[79.802 ; 82.251]
mIoU seed 42	81.814	81.561
mIoU seed 123	81.339	80.556
mIoU seed 456	81.701	81.388

$\Delta(\text{MoE} - \text{control}) = +0.449 \text{ pt} \cdot 95 \% \text{ CI } [+0.109 ; +0.820] \cdot p = 0.0066 \rightarrow \text{significant at the raw threshold.}$

The three multiplicity families, given together (sizes computed from the artifacts, Holm recomputed from the raw p):

Family	Size	Holm of the primary	Source artifact
pre-registered single pair (P3.10)	1	0.0066 (= p)	<code>harness/table_controle_v3cs_P310.</code>
arms against control (P3.14)	12	0.0726	<code>p314/master_table.json</code>
program exploratory (P3.16)	15	0.0924	<code>metiers_experts/table_metiers_exp</code>

Honest reading: the primary endpoint is significant at the raw threshold ($p = 0.0066$) and **inside the pre-registered 1-pair family** (Holm = 0.0066). It **does not survive the exploratory 15-pair family** (Holm = 0.0924 $>$ 0.05) — and **no arm of the plateau survives it either** (best Holm 0.0750). Stated as-is: on mIoU, the conclusions of this program rest on **amplitudes and their CIs**, and the contribution of this paper is what each arm **breaks** (§6.3), not the post-Holm significance of the mIoU.

Inter-forward robustness. A second independent GPU forward of the same checkpoints (business-metric regeneration path, §6.4) gives $\Delta = +0.450218 \text{ pt}$ against +0.449360 pt for the harness — a gap of **8.58×10^{-4} pt** (6.07×10^{-4} pt on the MoE mIoU, 2.51×10^{-4} pt on the control mIoU), i.e. cuDNN/BF16 non-determinism. Same verdict in both sources.

Δ mIoU des 13 bras du programme (référence : contrôle apparié, 80 époques)
IC95 bootstrap apparié B = 10,000. Aucun bras ne survit au Holm : meilleur 0.0576 sur 12 paires, 0.0750 sur 15 paires

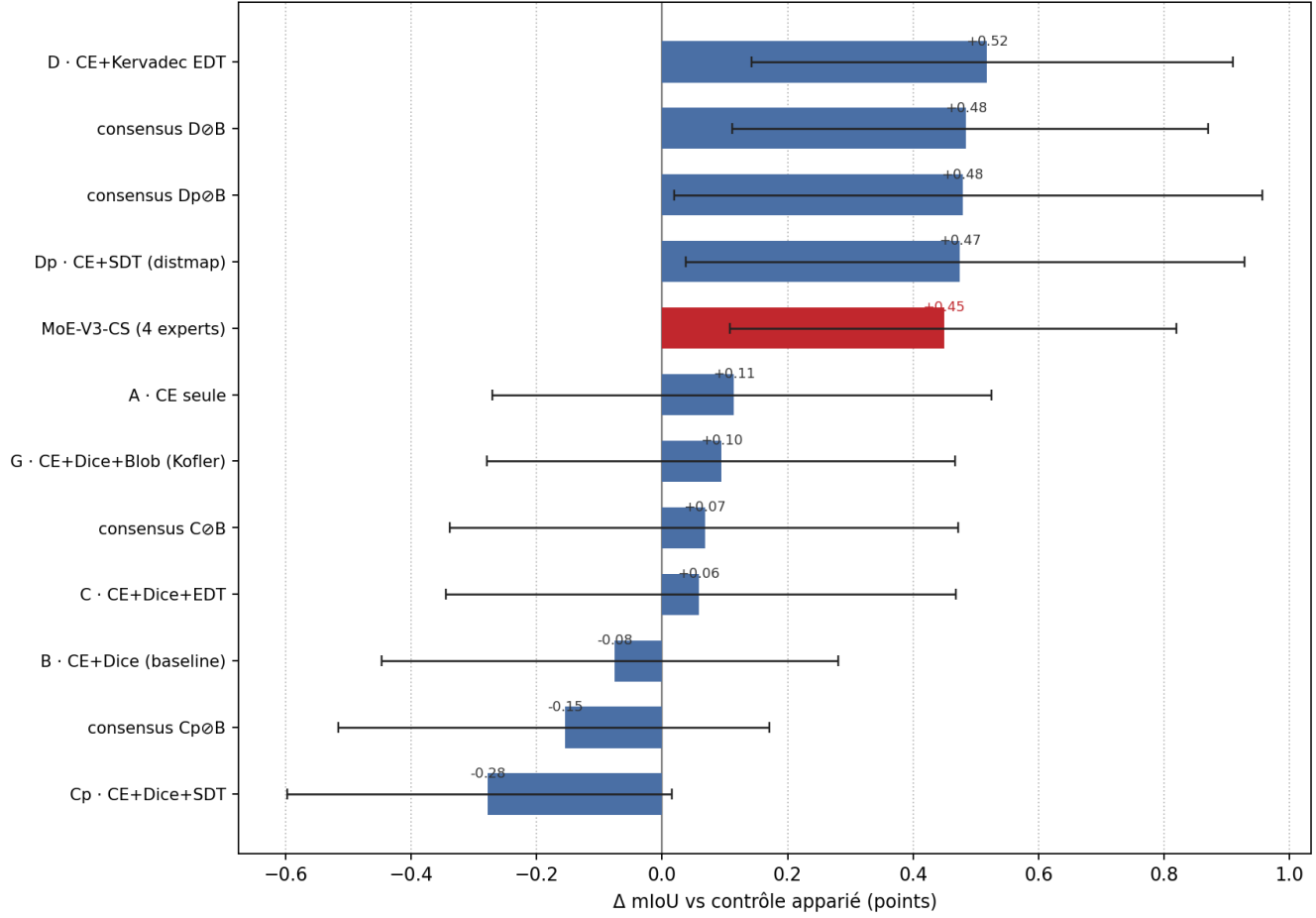


Figure 1: F1 — Δ mIoU of the 13 arms against the matched control, with 95 % CI

Figure 1: Δ mIoU vs the matched control for the 13 arms of the program (paired bootstrap, B = 10 000). MoE-V3-CS is 5th by amplitude; the two families of Holm are both reported in Table T3.

Position among the 13 arms (reference: matched control; artifact results/moe_v3_cs/p314/master_table.json). Both Holm families are given, with their size computed from the artifact:

#	Arm	mIoU	Δ vs control	95 % CI	p	Holm (12 pairs, P3.14)	Holm (15 pairs, P3.16)
1	D · CE+Kervadec EDT	81.686	+0.518	[+0.142 ; +0.910]	0.0048	0.0576	0.0750
2	consensus DøB	81.652	+0.484	[+0.111 ; +0.871]	0.0082	0.0820	0.1066
3	consensus DpøB	81.647	+0.479	[+0.019 ; +0.957]	0.0396	0.3168	0.4136
4	Dp · CE+SDT (dismap)	81.643	+0.475	[+0.038 ; +0.927]	0.0312	0.2808	0.3960
5	MoE-V3-CS (4 experts) ← this paper	81.618	+0.449	[+0.109 ; +0.820]	0.0066	0.0726	0.0924

#	Arm	mIoU	Δ vs control	95 % CI	p	Holm (12 pairs, P3.14)	Holm (15 pairs, P3.16)
6	A · CE alone	81.283	+0.115	[−0.271 ; +0.525]	0.5722	1.0000	n.a. (a)
7	G · CE+Dice+Blob (Kofler)	81.263	+0.095	[−0.279 ; +0.467]	0.6552	1.0000	1.0000
8	consensus C $\oslash$ B	81.236	+0.068	[−0.338 ; +0.472]	0.7758	1.0000	1.0000
9	C · CE+Dice+EDT	81.227	+0.059	[−0.344 ; +0.468]	0.8096	1.0000	1.0000
10	B · CE+Dice (baseline)	81.093	−0.075	[−0.447 ; +0.281]	0.6748	1.0000	1.0000
11	consensus Cp $\oslash$ B	81.014	−0.154	[−0.516 ; +0.170]	0.3498	1.0000	1.0000
12	Cp · CE+Dice+SDT	80.890	−0.278	[−0.597 ; +0.016]	0.0628	0.4396	0.6180
—	control (CE+Dice 80 ep, ref)	81.168	0 (ref)	—	—	—	—

(a) A is outside the 15-pair family (partial coverage). The MoE-V3-CS is **5th of 12** by amplitude (5th of 13 with the control, which at $\Delta = 0$ ranks 10th). **5 arms have a raw $p < 0.05$; none survives Holm**, neither over 12 pairs (best 0.0576) nor over 15 (best 0.0750). This is the paper’s pivot: since the mIoU amplitude does not separate the arms significantly after correction, the conclusion rests on **what each arm breaks** (§6.3), where MoE-V3-CS is first by a measured margin.

5.2 Routing diagnostic: the gate does not choose

Read from `routing_moe_v3cs_Binit_seed*.jsonl` (80 epochs $\times$ 3 seeds), cross-checked against the summaries’ `final_epoch_stats` (identity verified to 1×10^{-12} on 8 quantities $\times$ 3 seeds). Exact equidistribution of `part_max` for top-2 of 4 experts = **0.5000** (computed `top_k / n_experts`).

Quantity	epoch 0 (seed 42/123/456)	final epoch (seed 42/123/456)	reference	reading
entropy_norm	0.99970 / 0.99967 / 0.99955	1.00000 / 0.99771 / 0.99997	1.0 = uniform	near-uniform gate
part_max	0.50056 / 0.50149 / 0.50149	0.5030 / 0.5019 / 0.5019	0.5000 = equidistribution	max gap 0.0030
frac of the 4 experts dead experts (frac = 0)	—	[0.4979 ; 0.5030] 0	0.5000 0	no favoured expert no collapse
top_expert_share	0.66700 / 0.66898 / 0.65957	0.59755 / 0.59546 / 0.59695	1.0 = single expert	mean of per-batch maxima, quasi-stable
noise_std	1.00 / 1.00 / 1.00	0.00 / 0.00 / 0.00	annealed over 40 epochs	exploration off, flatness remains
γ (gamma_absmean)	0.00024 / 0.00023 / 0.00023	0.02026 / 0.01912 / 0.01815	0 = useless MoE	strictly positive, $\times 77$ to $\times 84$

Reading. At the final epoch the routing distribution is flat at a minimal normalised entropy of 99.771 %, `part_max` is 0.5030 / 0.5019 / 0.5019 against 0.5000 for exact equidistribution, the four experts receive between 49.79 % and 50.30 % of the patches, and **no expert is dead**. The gate therefore does not choose: it averages.

This is not an artifact of the Shazeer noise — and the proof is not the one one expects. At the final epoch the noise is annealed to 0, so `entropy_token_clean` is *identical by construction* to `entropy_token`: that particular equality proves nothing (it is a tautology). The **probative** comparison runs over the **40 epochs where the noise is active** (std $1.0 \times$ std of logits): there the maximal gap between the two entropies is 1.45×10^{-3} (seed 42 9.54×10^{-4} , seed 123 9.95×10^{-4} , seed 456 1.45×10^{-3}), while the

noise-free distribution stays flat at ≥ 0.99821 normalised entropy and `part_max` stays within **[0.50011 ; 0.50217]** of the 0.5000 equidistribution. In other words, the flatness of the routing is a property of the *gate*, not of the noise injected into it.

La porte ne choisit pas — 80 époques $\times$ 3 seeds
 Zone rose : les 40 époques où le bruit de Shazeer est ACTIF (seule phase où la comparaison avec/sans bruit est probante ; après, le bruit est nul et les deux courbes sont identiques par construction)
 Époque finale : `entropy_norm` 0.99771 au minimum, `part_max` 0.5019–0.5030 pour 0.5000, 0 expert mort, $|\gamma|$ 0.01815–0.02026

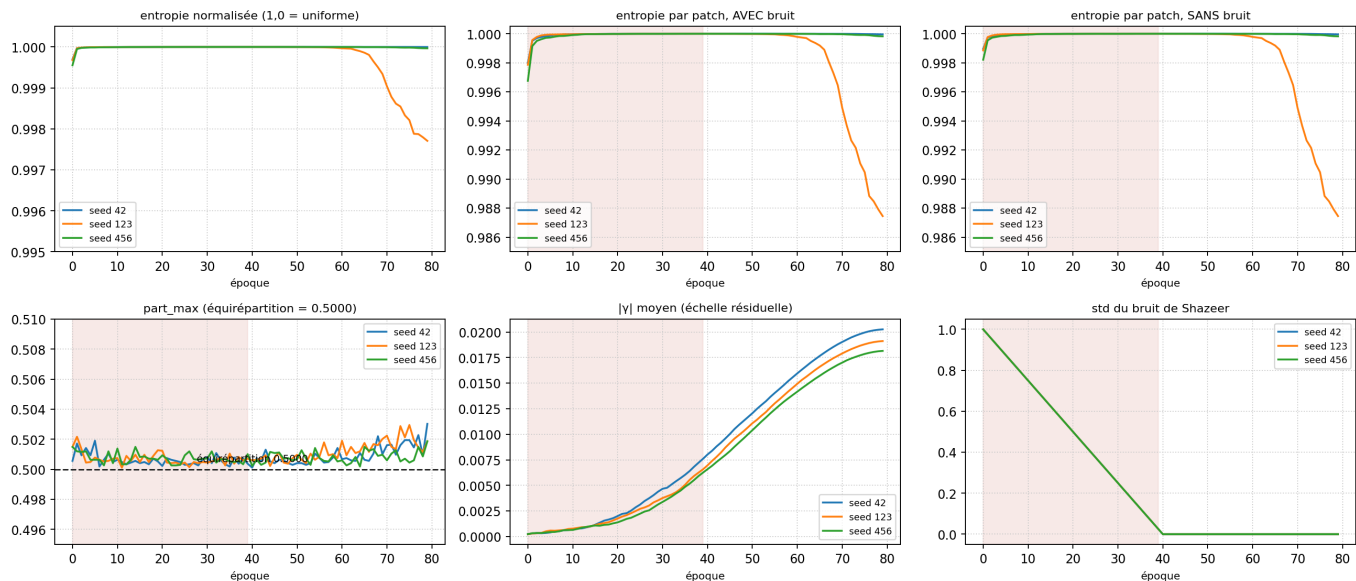


Figure 2: F2 — Routing over 80 epochs $\times$ 3 seeds: frac, entropies, γ

Figure 2: routing diagnostics over training. Top: per-expert *frac* stays pinned at 0.5000 (equidistribution) with no dead expert. Middle: the noise-free entropy stays ≥ 0.99821 during the 40 active-noise epochs. Bottom: γ grows from 0.00024 to ≈ 0.020 — the mixture helps, but as an average.

Yet the mixture **helps**: γ , the learnable per-channel residual scale (initialised to 0.0 to make epoch 0 bit-exact to the control), reaches 0.02026 / 0.01912 / 0.01815 at the end of the run against 0.00024 / 0.00023 / 0.00023 at epoch 0 ($\times 77$ to $\times 84$). The gain therefore comes from the **average of experts**, not from selection — the replication, on Cityscapes, of the BRATS result, with another architecture and another attach point.

Limitation, stated as-is. `entropy_token` of seed 123 (0.98745) is very slightly below the other two (0.99996, 0.99982): the thesis holds over the 3 seeds but is not of perfect uniformity on that one.

5.3 Business metrics: what the mixture captures, what it pays

Holdout first:500, seeds averaged within each replicate, paired bootstrap $B = 10\,000$ (seed 20260618), Holm over the **15-pair** family. Units: points ($\times 100$), except `fragments` = connected components per image. **Bold = Holm < 0.05** . The two criteria (Holm and CI-excluding-zero) are separated, never merged.

Metric	n	control	MoE-V3- CS	$\Delta(\text{MoE-ctl})$	95 % CI	p	Holm(15)	verdict
mIoU	500	81.168	81.618	+0.450	[+0.110 ; +0.818]	0.0066	0.0924	CI excludes 0, does not survive Holm
boundary_f500px	500	76.627	76.567	-0.061	[-0.178 ; +0.055]	0.3004	1	ns

Metric	n	control	MoE-V3- CS	$\Delta(\text{MoE-ctl})$	95 % CI	p	Holm(15)	verdict
rappel_str_141_instances	76	76.253	76.728	+0.475	[−0.004 ; +0.935]	0.0516	0.2064	ns
precision_173_pixels	70	70.028	70.024	−0.004	[−0.377 ; +0.376]	0.9896	0.9896	ns
instances_441tes_rappel	79	80.1975	80.372	+0.397	[−0.028 ; +0.766]	0.0694	0.2776	ns
instances_440individueltes_rappel	80	80.096	80.504	+0.407	[−0.030 ; +0.783]	0.0662	0.2648	ns
instances_63rule_rappel	74	74.584	74.199	−0.384	[−1.536 ; +0.493]	0.4822	1	ns
instances_845ille_T1_rappel	64	64.994	65.796	+0.802	[+0.073 ; +1.510]	0.0330	0.198	CI excludes 0, does not survive Holm
instances_835ille_T3_rappel	93	93.474	93.733	+0.258	[+0.142 ; +0.384]	0.0000	0	Holm-significant
IoU_person500	85	85.152	84.910	−0.242	[−0.429 ; −0.062]	0.0080	0.088	CI excludes 0, does not survive Holm
IoU_rider 500	68	68.920	68.555	−0.365	[−1.302 ; +0.524]	0.4298	1	ns
fragments 500	519	519.3	488.0	−31.2	[−35.7 ; −26.7]	0.0000	0	Holm-significant

(full 20-metric table in tables/T5_metiers.md). Over the 20 business metrics, **2 survive the Holm** on the 15-pair family — **instances_taille_T3_rappel** (+0.26 pt, largest-instance recall) and **fragments** (−31.2 components per image), both with $p = 0$ — and **5 have a CI excluding zero**. It is tempting to write that only the fragments survive; that is false, and the second survivor goes *in the direction of the thesis*: the only business gain that resists multiplicity is on the **large** instances (T3), while the gain on the **small** ones (T1 +0.80) does not resist (Holm 0.198). The mixture improves what is already well seen and reduces fragmentation; it does not repair the smallest instances.

Métriques métier : 2 des 20 survivent au Holm (`instances\_taille\_T3\_rappel`, `fragments`). Les deux critères de significativité sont tracés séparément, jamais confondus.

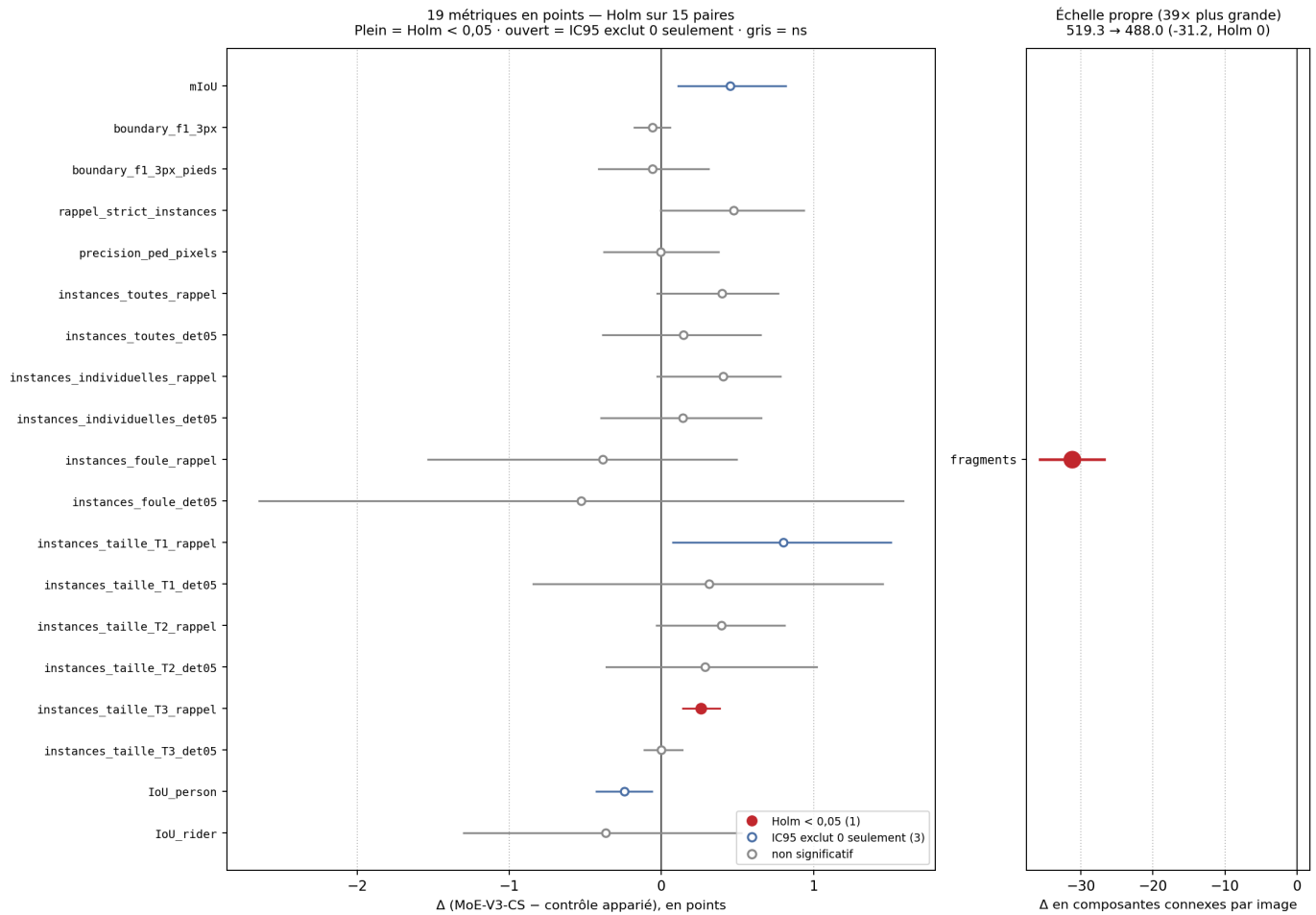


Figure 3: F3 — Business-metric forest plot, two panels at distinct units

Figure 3: business-metric $\Delta(\text{MoE} - \text{control})$ with 95 % CI. The two panels carry distinct units — the 19 point-valued metrics on one axis, *fragments* ($\Delta -31.2$ components per image) on its own — because plotting them together would compress the point metrics by a factor of ≈ 39 into an unreadable band.

What the mixture captures from the experts, and what it does not inherit (Δ vs control, same artifact):

Metric	Δ MoE	Δ D	Δ Dp	Δ G	computed reading
instances_taille_T3_rappel	+0.802	+2.816	+2.016	-4.080	captures part of D/Dp's gain
rappel_strict_instances	+0.475	+2.183	+1.719	-3.417	captures part of D/Dp's gain
precision_ped_pixels	+0.104	-22.195	-15.020	-6.763	loss, but 0.00 $\ll$ -22.20 (D)
instances_foule_rappel	+0.384	+1.511	+1.319	-4.612	loss, but 0.38 $\ll$ -4.61 (G)
traffic light (per-class)	-0.081	-2.602	-2.340	+0.470	does not inherit the boundary/distmap loss

Metric	Δ MoE	Δ D	Δ Dp	Δ G	computed reading
boundary_f1_3px	-0.061	+0.677	+0.545	+0.376	neutral

Calculated reading: MoE-V3-CS captures part of the instance gain of D/Dp (T1 +0.80 against +2.82 and +2.02) **without** inheriting either their traffic light loss (-0.08 against -2.60 and -2.34) or the pedestrian recall destroyed by G (+0.47 against -3.42). Its measured counterpart: **no contour gain** (boundary_f1_3px -0.061, ns) — consistent with §5.2, an average of experts smooths out the contour specialities. The worst family of the mixture is contours (-0.06 pt on average) and it is its **only** negative family (1 of 5).

5.4 Per-class IoU: three criteria, given separately

19 Cityscapes classes, reference = matched control, Holm over the **19-class family** (recomputed from the raw p, max gap 0.0). Bold = Holm(19) < 0.05. Over the 19 classes: **11 rise, 8 fall**.

Class	IoU MoE	IoU ctl	Δ (MoE-ctl)	95 % CI	p	Holm(19)	Δ D	Δ Dp	Δ G	Δ B
truck	84.834	80.593	+4.241	[+0.438 ; +8.752]	0.0066	0.1188	+4.94	+5.19	+3.43	-0.35
wall	58.698	56.147	+2.551	[-1.085 ; +6.897]	0.2000	1	+5.49	+3.08	+1.07	+1.77
train	83.463	82.234	+1.229	[+0.054 ; +2.820]	0.0408	0.5712	+0.40	+0.66	-1.04	+1.50
fence	66.683	65.807	+0.875	[-0.271 ; +2.298]	0.1586	1	+1.21	+1.06	-0.46	+0.26
sidewalk	87.296	86.975	+0.321	[+0.040 ; +0.660]	0.0224	0.336	+0.62	+0.48	+0.06	+0.09
road	98.468	98.411	+0.057	[+0.013 ; +0.111]	0.0078	0.1326	+0.05	+0.07	+0.04	+0.03
person	84.910	85.153	-0.244	[-0.433 ; -0.062]	0.0078	0.1326	-0.19	-0.25	+0.07	-0.26
bicycle	80.068	80.333	-0.265	[-0.408 ; -0.121]	0.0002	0.0038	-0.70	-0.53	+0.05	-0.52

(selected rows; full 19-class table in tables/T6_perclass.md). The ranking by **amplitude** and the ranking by **significance** do not coincide; conflating them would produce a false statement. The three criteria are given as-is:

1. **Largest amplitudes** (decreasing Δ): truck +4.24, wall +2.55, train +1.23, fence +0.88.
2. **CI excluding zero** (the P3.16 “significant” criterion): rises truck +4.24, train +1.23, sidewalk +0.32, road +0.06; falls person -0.24, bicycle -0.26.
3. **Holm over the 19-class family: no rise survives**; the only surviving class is a **fall** — bicycle (Holm 0.0038).

This is the central honesty point of the table: the most spectacular gain (truck +4.24 pt) has a CI excluding zero but **does not survive** the 19-class Holm (0.1188), because its CI is wide ([+0.44 ; +8.75]) on a rare class. The only per-class effect that resists multiple correction is a **cost** (bicycle), not a gain. The costs are small and distributed: bicycle -0.26, motorcycle -0.31, rider -0.37.

IoU par classe — trois critères donnés séparément
Plus fortes amplitudes : truck, wall, train, fence · IC95 excluant 0 : truck, train, sidewalk, road (hausse) et person, bicycle (baisse) · Holm(19) : AUCUNE hausse ne survit, la seule classe survivante est une baisse (bicycle, Holm 0.0038)

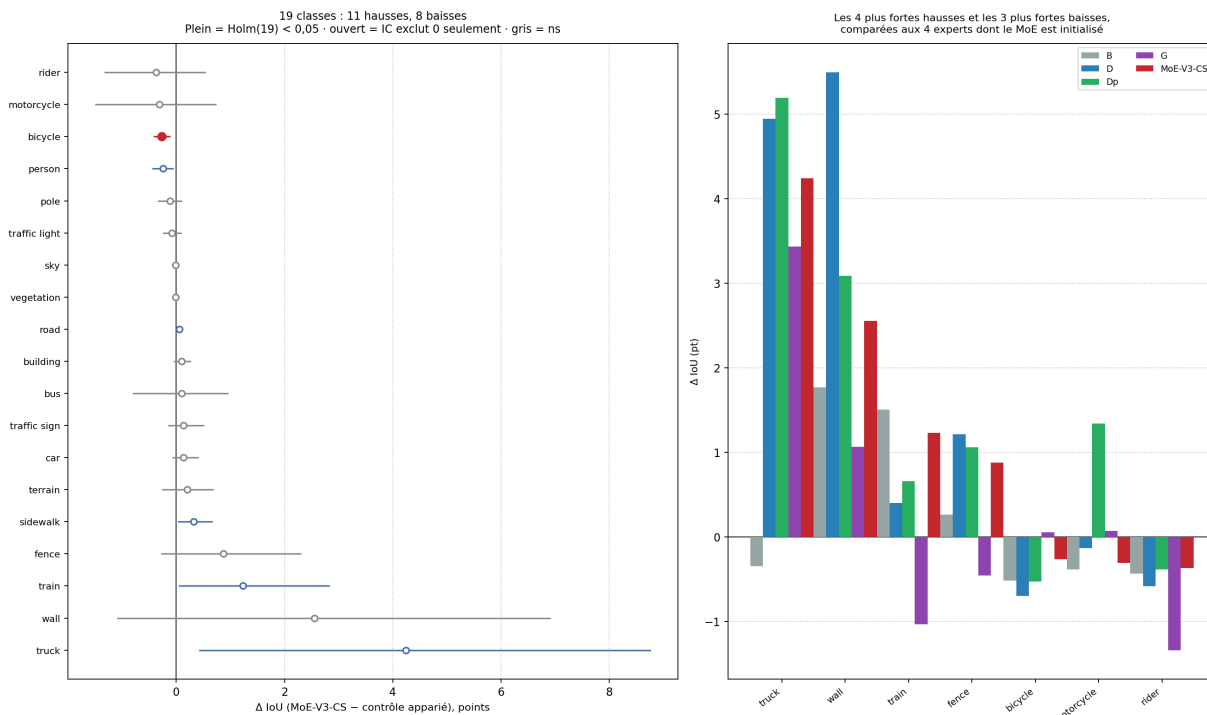


Figure 4: F5 — Per-class IoU heatmap, MoE vs control and vs the four specialists

Figure 5: per-class Δ IoU. The mixture recovers 82 % of the best specialist gain on truck ($Dp + 5.19$) and 82 % on train (where the best expert is $B + 1.50$, not D nor Dp — a denominator limited to D/Dp would have written “186 % of the best specialist gain”, which is meaningless), while keeping the traffic-light loss at 3 % of D ’s and the pedestrian loss at 95 % of B ’s.

5.5 Versatility: the only all-rounder, first on nothing

The program’s pre-registered versatility criterion: **36 endpoints** (19 per-class IoU + 17 business metrics) $\times$ 13 arms, same holdout and bootstrap. “Maximal damage” = the worst endpoint-level Δ vs control (closer to 0 = no weak point). The criteria T0–T4 were written **before** reading the results, and each verdict below is **recomputed here from the raw deltas and ranks**, then compared against the stored verdict — the thesis “only all-rounder arm” is re-demonstrated, not copied.

Criterion	Statement	Recomputed result	Verdict
T0	significant Δ IoU and none of the 36 endpoints more than 1 pt below reference and none of the 19 classes degraded more than 0.5 pt	1 arm of the 11 ranked against the reference: MoE-V3-CS	✓
T1	lowest maximal damage of the plateau	−0.53 pt (instances_foule_det05) against −1.90 (consensus C∅B) and −22.20 (D) — margin 1.37 pt	✓
T2	significant Δ IoU nonetheless	$\Delta +0.45$ pt, $p = 0.0066$, Holm(12) 0.0726, Holm(15) 0.0924	✓ raw, × after exploratory Holm — stated as-is

Criterion	Statement	Recomputed result	Verdict
T3	no endpoint won outright	0 rank-1 of 36 (D has 16), worst rank 11/12 on instances_foule_rappel, never last (0 last places)	✓
T4	verdict stable seed by seed	1st in damage on 2 of 3 seeds (42, 456; seed 123 → 3rd)	× partial

Maximal damage per arm — the ranking that founds the thesis (full table in tables/T7_polyvalence.md):

Rank	Arm	max damage (pt)	endpoint	z	Δ mIoU (pt)	p	price per mIoU pt	percentile mean	worst rank	#1
1	MoE-V3-CS (4 experts)	-0.53	instances_foule_det05	-0.35	+0.545	0.0066	-1.2	60.4	11/12	0
2	consensus C⊙B	-1.90	IoU_bus	-1.47	+0.07	0.7866	—	44.0	13/13	0
6	consensus Dp⊙B	-4.77	precision_232_pixels	-0.74	+0.48	0.0376	-9.9	69.0	12/12	4
9	consensus D⊙B	-8.70	precision_135_pixels	-1.35	+0.49	0.0082	-17.9	71.9	12/13	3
10	Dp · CE+SDT (distmap)	-15.02	precision_232_pixels	-2.32	+0.47	0.0330	-31.9	72.7	13/13	3
11	D · CE+Kervadec EDT	-22.20	precision_344_pixels	-3.44	+0.52	0.0050	-43.1	77.3	13/13	16

“Price per mIoU point” = maximal damage divided by Δ mIoU, for the arms whose Δ mIoU is significant at the raw threshold: it is the exchange rate between the global gain and what the arm breaks. MoE-V3-CS is the cheapest of the plateau by a measured factor (-1.2 against -43.1 for D).

Every rank carries the denominator of its own endpoint. The worst rank of the mixture is **11/12**, reached on instances_foule_rappel and instances_foule_det05. The denominator is **not constant** (it is not always 13): arm A is in partial coverage (annexe_couverture_partielle of P3.17), ranked on only **20** of the 36 endpoints, so the other 16 rank only **12** arms. “Never last” stays true (11 < 12, the last place there is 12), but with the right denominator.

Mean level: “good everywhere” ≠ “best everywhere”. In mean percentile, D (77.3) > Dp (72.7) > consensus D⊙B (71.9) > **MoE-V3-CS (60.4, 5th of 11)**. On a single target the specialist stays ahead. The mixture is first on **none** of the 36 endpoints (D wins 16). What it is the only one to do is to be **nowhere broken**: maximal damage -0.53 pt where D goes down to -22.20 pt.

Two counters, not to be confused (both computed): **22 of the 36 endpoints are above the control** ($\Delta > 0$; 14 below, none null) and **28 are in the upper half of the ranking** among the 12 arms (8 in the lower half). The second measures not a sign but a **relative position**: an arm can be in the upper half while staying below the control, and vice versa.

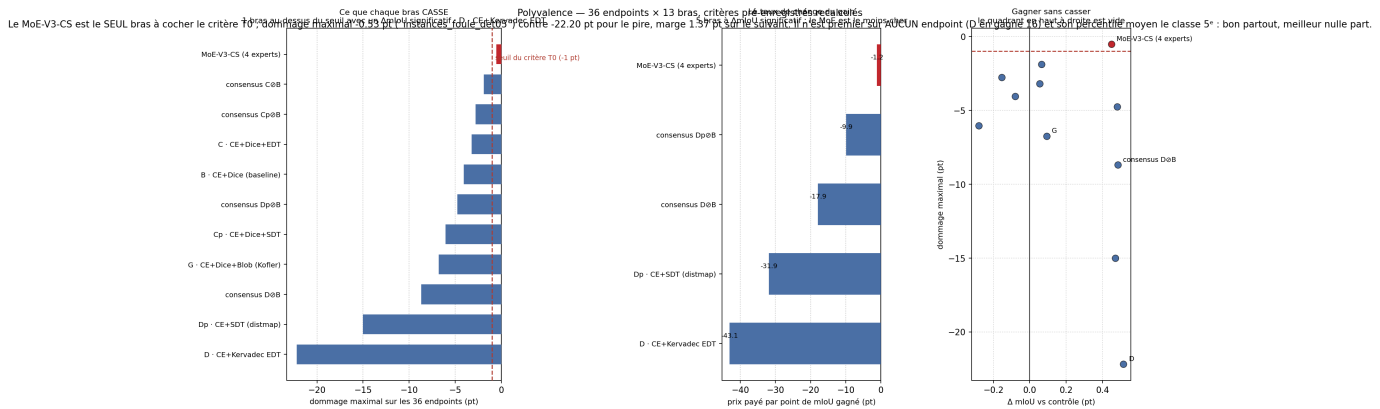


Figure 5: F4 — Versatility: 4 panels (damage, percentile, seed stability, families)

Figure 4: the versatility verdict. The mixture is first in maximal damage (-0.53 pt, margin 1.37 pt) yet 5th in mean percentile (60.4) and first on 0 of 36 endpoints — the all-rounder is not the best.

Seed-by-seed stability. The mixture is the least damaged arm on 2 of 3 seeds (42, 456); seed 123 puts it 3rd behind consensus C $\odot$ B and Cp $\odot$ B — but those two arms have a Δ mIoU of $+0.07$ and -0.15 pt (non-significant): they break nothing because they do nothing. That is why T4 is declared **partial** rather than passed.

6. Discussion

6.1 Why an average of experts beats a selection

The gate stays flat ($\$5.2$), yet the mixture beats its matched control ($+0.449$ pt, $p = 0.0066$) and γ grows to ≈ 0.020 . The reading the measurements support is that the win comes from the **expert initialisation**, not from the routing: starting from four differentiated specialists (relative Frobenius distance median 0.538 across methods) and averaging them with a learned per-channel residual scale produces a function that is better than the control *everywhere on average* without ever committing to one expert per patch. This is the same mechanism the BRATS track reported [DOI 10.5281/zenodo.22903668], on a different architecture and attach point — which is the generality argument of this paper. The flatness is a property of the gate, not of the injected noise: during the 40 active-noise epochs the noise-free distribution is already flat (≥ 0.99821) and `part_max` stays within 0.00011 – 0.00217 of equidistribution.

6.2 The residual guard γ is an architectural fact, not a tuned knob

γ was **not** necessary on BRATS (identical MedNeXt blocks everywhere + GroupNorm: the experts saw, at the attach point, a distribution close to the one they were trained on). It is **indispensable** here: without it the raw recipe costs -28.1 mIoU points at epoch 0, because the UPerNet laterals + BatchNorm present the experts, at `head.fpn_bottleneck`, with a distribution they have never processed and no renormalisation before the classifier. Initialising γ to 0 makes epoch 0 bit-exact to the control, and turns $|\gamma|$ into a direct diagnostic of the thesis: $|\gamma| \rightarrow 0$ would mean the mixture is useless; $|\gamma| \approx 0.020$ means it contributes. This is a **measured** difference between the two datasets, not a post-hoc setting.

6.3 Why no per-class gain survives Holm while the mixture still wins

The only per-class effect that resists the 19-class Holm is a **cost** (bicycle, -0.26 , Holm 0.0038); the largest gain (truck $+4.24$) has a wide CI on a rare class and does not survive. This is not a contradiction with the positive primary: the mIoU is the *mean over 19 classes*, and the mixture raises the mean by spreading small gains over many classes (11 rises) while keeping the falls small and distributed (worst -0.37 on rider). A per-class Holm is underpowered for exactly this kind of distributed effect — which is why the program’s conclusions rest on amplitudes and CIs, and on the versatility criterion ($\$5.5$) rather than on per-class significance.

6.4 Relation to the consensus arms

Two training-free consensus arms (D $\odot$ B $+0.484$, Dp $\odot$ B $+0.479$) reach a slightly higher mIoU amplitude than the mixture ($+0.449$), and one (D $\odot$ B) has a higher mean percentile (71.9 vs 60.4). But each consensus arm carries a maximal damage of -8.70 and -4.77

pt (pedestrian pixel precision), against -0.53 pt for the mixture, and each wins outright on 3 and 4 endpoints respectively while the mixture wins on none. The mixture is not the strongest arm; it is the **only one that is nowhere broken**, which is the property the pre-registered T0 criterion was written to single out.

7. Limitations

1. **Holdout first:500.** A pre-specified subset of val, identical for all 13 arms and fixed before training — but *not* a random draw, and the official leaderboard was not submitted to.
 2. **No arm significant on mIoU after Holm.** Best raw $p = 0.0048 \rightarrow$ Holm 0.0576 over the 12-pair family, 0.0750 over the 15-pair exploratory family. Program-wide conclusions on mIoU rest on amplitudes and CIs, stated as such; the primary survives only the pre-registered 1-pair family.
 3. **One attach point, one MoE configuration.** `head.fpn_bottleneck` only; 4 experts, top-2, 3×3 grid only. No sweep over the number of experts, the attach point, top_k or the grid.
 4. **Cost.** The four experts must exist (4×160 epochs) *plus* 80 epochs of mixture and 80 of control; the mixture costs **+24.7 % time per epoch** (762 s vs 611 s) and **+3.7 GB VRAM** (35.2 vs 31.5 GB) over the control, ≈ 17.0 h per seed.
 5. **Control matched in recipe, not in total budget.** The control trains 80 epochs (the mixture’s recipe); a 160-epoch control was not trained. The comparison is paired in *recipe*, not in total compute.
 6. **Seed 123 slightly less uniform.** `entropy_token` 0.98745 against 0.99996 / 0.99982; T4 (seed-by-seed stability) is **partial** (1st in damage on 2 of 3 seeds), not passed.
 7. **$n = 3$ seeds.** Sub-0.5-pt effects are underpowered at the seed level; the primary inference is the 500-image paired bootstrap, which probes evaluation-set sampling, not seed variance.
 8. **cuDNN/BF16 non-determinism declared.** Two independent forwards of the same checkpoints differ by 8.58×10^{-4} pt on the primary Δ (verdicts identical); bit-exact GPU reruns are not claimed.
 9. **Versatility artifact mixes two forwards.** Its per-class columns come from the P3.16 attribution ($\Delta +0.449360$) and its mIoU column from the business regeneration ($\Delta +0.450218$); the identity “mean of the 19 per-class $\Delta == \Delta \text{mIoU}$ ” is exact *within* an artifact (verified, gap 0) but only to 8.58×10^{-4} pt across the two — the same inter-forward gap declared above, not an inconsistency.
-

8. Conclusion

Insert a patch-wise mixture into a full-resolution Cityscapes segmenter, initialise its four experts bit-for-bit from four loss specialists of the same seed, train the gate and a zero-initialised residual scale for 80 epochs, and the result is: **the mixture beats its recipe-matched control** ($\Delta \text{mIoU} +0.449$ pt [$+0.109$; $+0.820$], $p = 0.0066$, Holm = 0.0066 in the pre-registered 1-pair family; it does not survive the exploratory 15-pair family, Holm = 0.0924, where no arm of the plateau survives), **and its gate does not choose** — flat routing at the final epoch (minimal normalised entropy 0.99771, `part_max` within 0.0030 of the 0.5000 equidistribution, zero dead experts), a flatness that is a property of the gate and not of the annealed noise (over the 40 active-noise epochs the noise-free distribution stays ≥ 0.99821), while γ grows from 0.00024 to ≈ 0.020 . The gain comes from the **average of experts**, not from selection — the replication, on a second dataset, a second architecture and a second attach point, of the BRATS result *The Gate Does Not Choose* [DOI 10.5281/zenodo.22903668], of which this paper reuses no number. On the program’s 36-endpoint versatility criterion the mixture is the **only all-rounder** (maximal damage -0.53 pt, margin 1.37 pt, price -1.2 pt per mIoU point against -43.1 for D) while being **first on none** of the 36 endpoints and 5th in mean percentile: good everywhere is not best everywhere. The gate still does not choose — and that is exactly why the mixture is the only arm that is nowhere broken.

Code, configs, per-arm artifact pointers, tables, figures and regeneration scripts: github.com/guillaume-cassez/cityscape-moe-experts (release bundle of this paper). Program companions: Cityscapes distmap [DOI 10.5281/zenodo.21006236], Cityscapes boundary ablation [DOI 10.5281/zenodo.21006393], Cityscapes blob loss [DOI 10.5281/zenodo.23083560], BRATS MoE-V3 (anteriority) [DOI 10.5281/zenodo.22903668; concept 10.5281/zenodo.22776410].

Appendix A — Runtime and reproducibility

Stage	Hardware	Time	Output
4 experts (160 ep each, 3 seeds)	1 × RTX PRO 6000 96 GB Blackwell	pre-existing (companion arms)	head.fpn_convs.0 of B, D, Dp, G
Mixture, 80 ep, 1 seed	same	17.0 h (762 s/ep, 35.2 GB)	checkpoints/moe_v3cs_Binit_seed{s}/epoch_080.p
Control, 80 ep, 1 seed	same	≈ 13.6 h (611 s/ep, 31.5 GB)	checkpoints/control_nomoe_Binit_seed{s}/epoch_
Primary bootstrap (harness)	CPU	minutes	results/moe_v3_cs/harness/table_controle_v3cs_
Routing diagnostic	CPU	seconds	routing_moe_v3cs_Binit_seed*.jsonl (80 ep × 3 seeds)
Tables + figures of this paper	CPU (no GPU)	≈ 1 s	papers/paper3/{tables,figures}/

Reproducibility seeds. Training seeds 42, 123, 456 are set globally (PyTorch, NumPy, Python random, CUDA). The bootstrap seed is fixed (20260618, B = 10 000) and shared by every table of the program, so all CIs and p-values are exactly re-derivable from the released per-image artifacts. cuDNN benchmark is left **on** (`deterministic: false`) for training speed; bit-exact GPU reruns are therefore not claimed — the declared reproducibility statement is: same artifacts → same statistics bit-for-bit (verified, 257 blocking checks in `tables/SANITY.md`), and independent forwards of the same checkpoints → agreement within 8.58×10^{-4} pt on the primary Δ (measured). The tables and figures of this paper regenerate **without GPU** from the consolidated asserted source `tables/paper3_tables.json` (`scripts/p3_moe_tables.py`, `scripts/p3_moe_figures.py`).

References

- Cordts *et al.* (2016). *The Cityscapes dataset for semantic urban scene understanding*. CVPR.
 - Cassez (2026a). *Distance-Map Auxiliary Regression for Full-Resolution Cityscapes Segmentation*. Zenodo. DOI 10.5281/zenodo.21006236.
 - Cassez (2026b). *Boundary Loss Ablation for Full-Resolution Cityscapes Segmentation: When Dice Helps and When It Doesn't*. Zenodo. DOI 10.5281/zenodo.21006393.
 - Cassez (2026c). *Equal Weight per Instance Does Not Pay Alone: A Blob-Loss Auxiliary on Full-Resolution Cityscapes*. Zenodo. DOI 10.5281/zenodo.23083560.
 - Cassez & Larnier (2026d). *The Gate Does Not Choose: An Expert-Initialised Mixture-of-Experts Outperforms 24 Arms on BRATS 2023*. Zenodo. DOI 10.5281/zenodo.22903668 (concept 10.5281/zenodo.22776410).
 - Cassez & Larnier (2026e). *Two Models That Agree Beat the Best of Them Alone: Parameter-Free Connected-Component Consensus (BRATS)*. Zenodo. DOI 10.5281/zenodo.22904810.
 - Kervadec *et al.* (2019). *Boundary loss for highly unbalanced segmentation*. MIDL. arXiv:1812.07032.
 - Kofler *et al.* (2023). *Blob loss for biomedical image segmentation*. IPMI. arXiv:2205.08209.
 - Lepikhin *et al.* (2020). *GShard: Scaling giant models with conditional computation and automatic sharding*. ICLR 2021. arXiv:2006.16668.
 - Perazzi *et al.* (2016). *A benchmark dataset and evaluation methodology for video object segmentation*. CVPR.
 - Shazeer *et al.* (2017). *Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer*. ICLR. arXiv:1701.06538.
 - Woo *et al.* (2023). *ConvNeXt V2: co-designing and scaling ConvNets with masked autoencoders*. CVPR. arXiv:2301.00808.
 - Xiao *et al.* (2018). *Unified perceptual parsing for scene understanding*. ECCV. arXiv:1807.10221.
-